Google Certificate Data Analyst Capstone

Project: Bellabeat Fitness Behavior Analysis

Data Analysis Portfolio Case Study

Python + Tableau/Spreadsheets

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Data Source & Preparation

- Fitbit activity dataset used for behavioral analysis
- Data cleaned and merged using Python (Pandas)
- Aggregated daily metrics for steps, calories, activity intensity, and sleep
- Processed dataset exported to Tableau for visualization

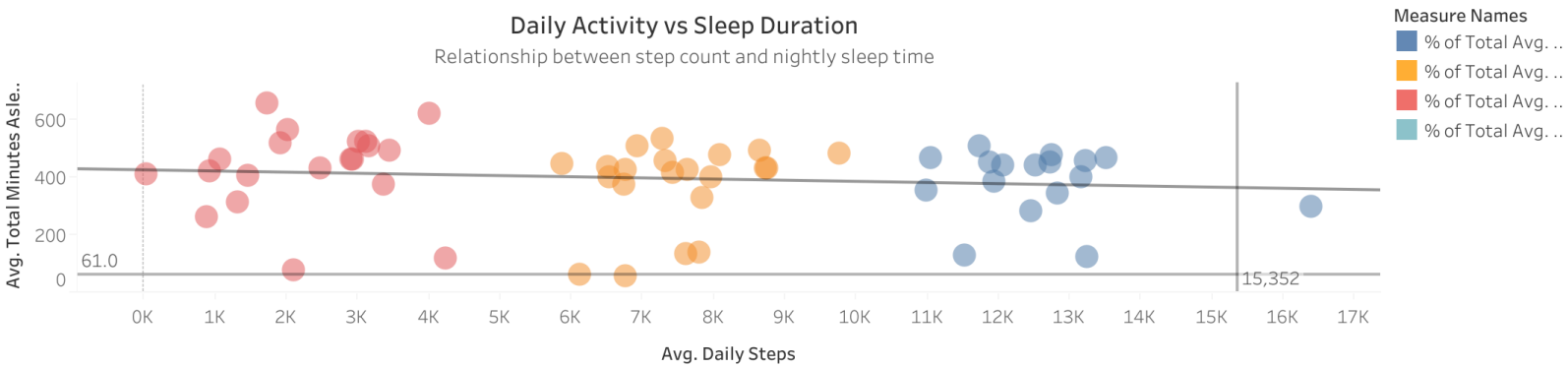
Python WorkFlow

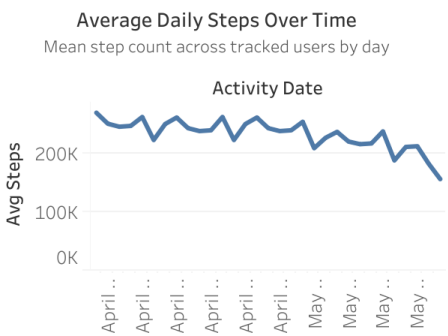
- Loaded activity and sleep datasets using Pandas
- Cleaned and standardized date formats
- Merged activity and sleep datasets into a unified dataframe
- Created day_of_week variable for behavioral analysis
- Calculated average activity levels and percent change metrics
- Exported processed dataset for Tableau dashboard analysis

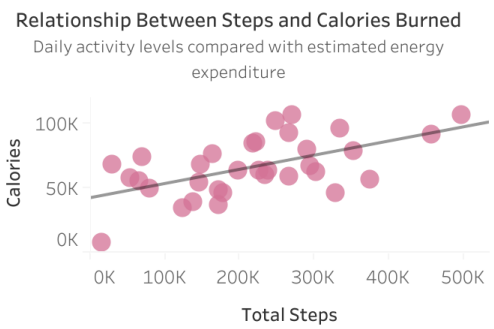
Data Limitations

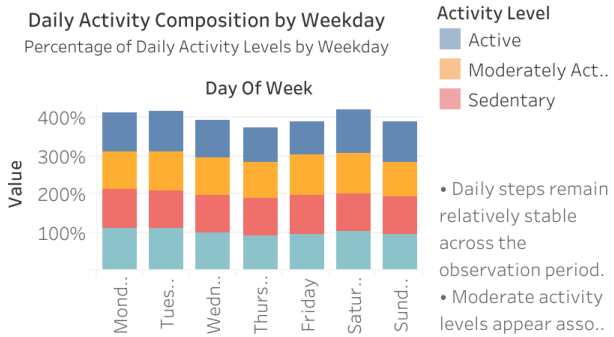
- Dataset includes only two months (March–April) of data
  Cannot assess long-term behavioral trends or seasonality.
- No demographic information available
  Cannot segment by age, gender, income, or geography.
- Sleep record was limited
- No yearly trends
- No holiday analysis
- No predictive modeling beyond short-term behavior

Bellabeat Fitness Behavioral Dashboard





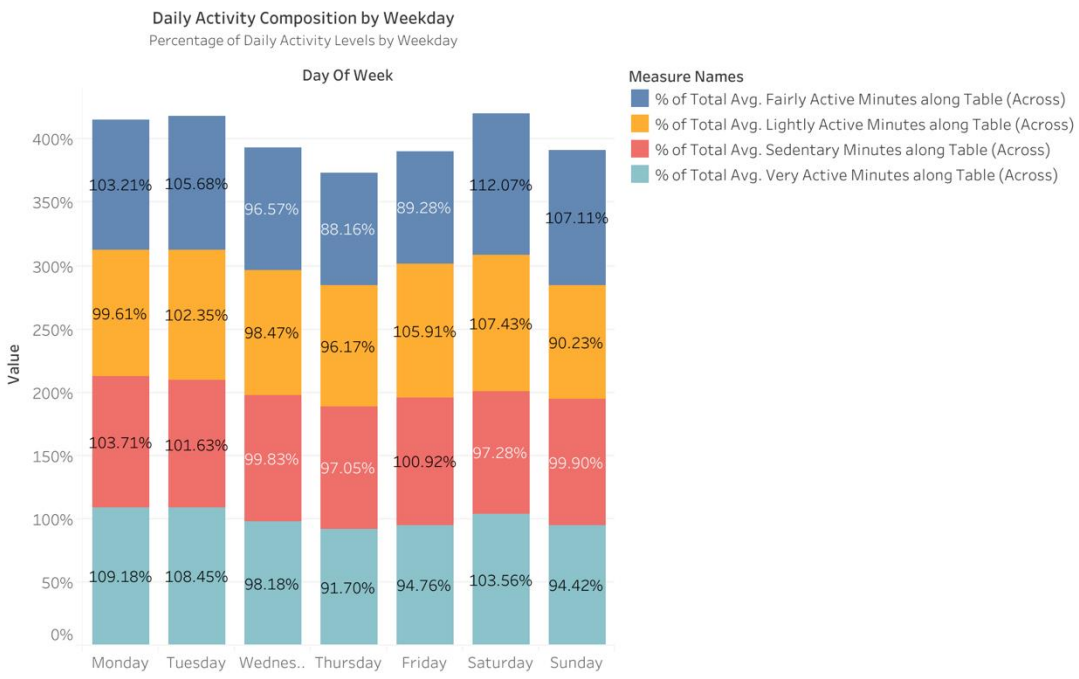




[Open Bellabeat Dashboard]

Daily Activity Composition by Weekday

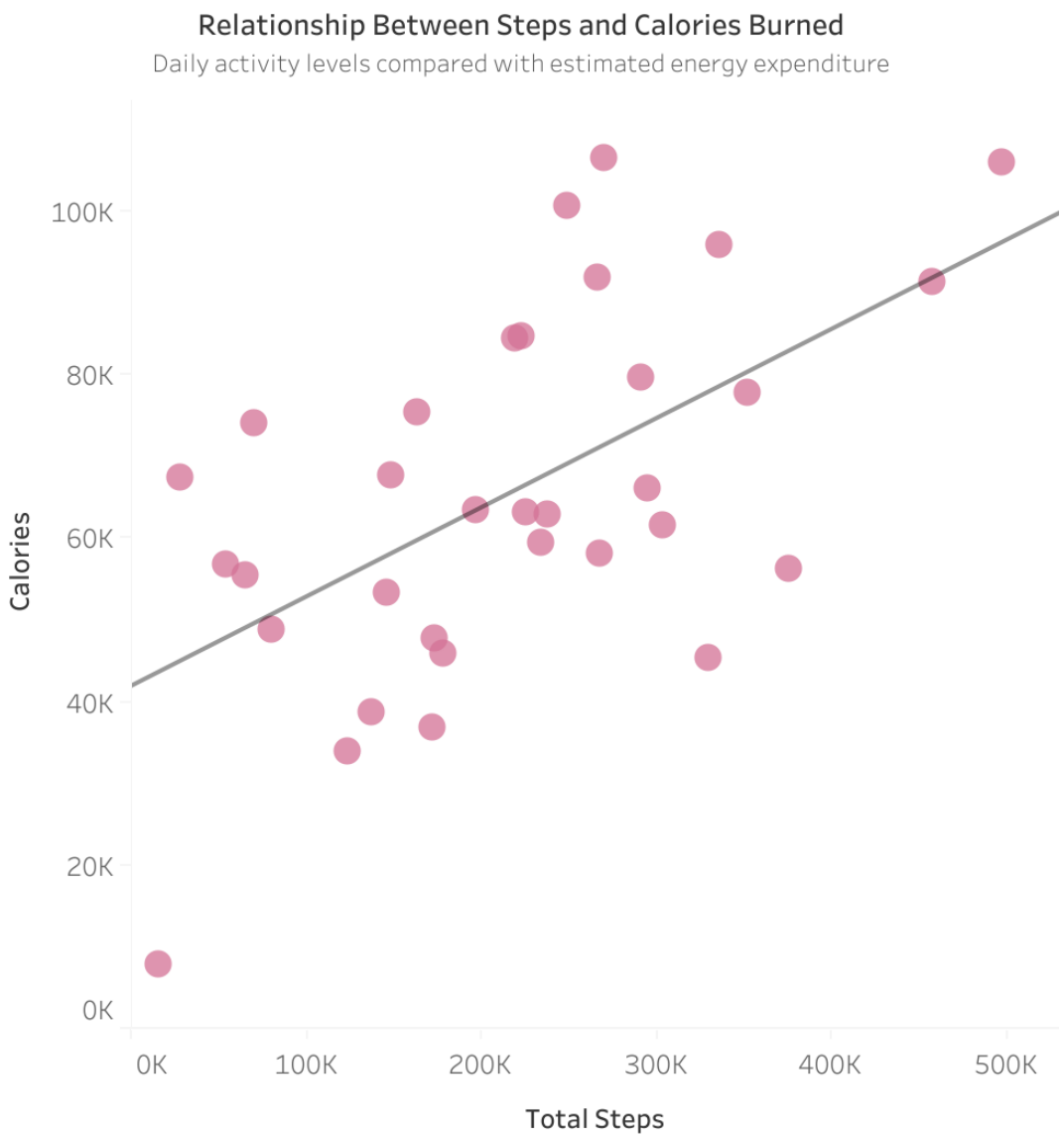
- Overall, activity composition remains relatively consistent across the week, with a noticeable increase in activity on Saturdays and a slight dip in intensity around Thursday and Sunday.



Relationship Between Steps and Calories Burned

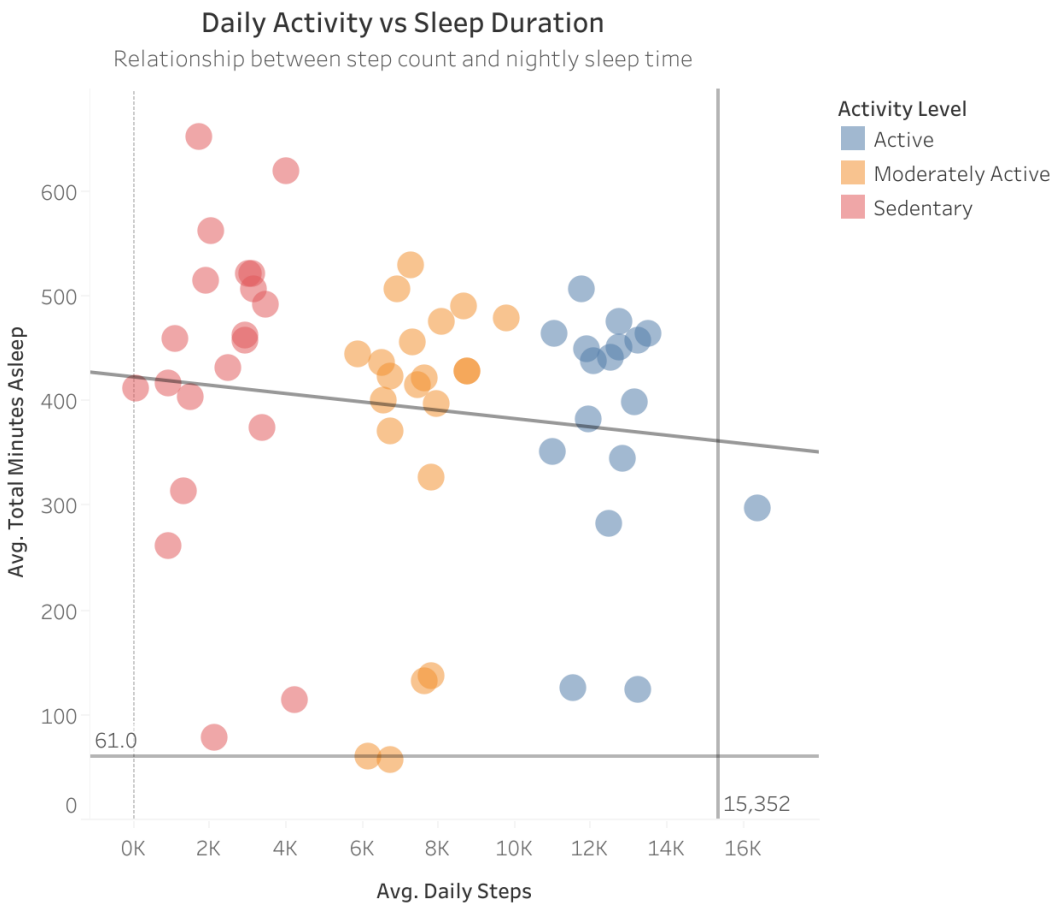
- This visualization shows a strong positive relationship between total steps and calories burned, confirming that higher levels of physical activity are associated with increased daily energy expenditure.

- These findings suggest that encouraging users to increase daily step counts could be an effective way to improve overall energy expenditure and promote healthier activity habits.



Daily Activity Vs Sleep Duration

- While the trend line suggests a slight negative relationship between step counts and sleep duration, moderately active users show the most consistent sleep patterns, generally clustering around 400–500 minutes per night. Sedentary users display the greatest variability in sleep duration, while highly active users do not consistently sleep longer despite higher step counts. These patterns suggest that balanced activity levels may be associated with more stable sleep routines.



Key Findings

- Daily step counts remain relatively stable across the observed period.
- Increased step counts strongly correlate with calories burned.
- Moderate activity levels appear associated with higher average sleep duration.
- Activity levels appear slightly lower on Sunday
- Saturday activity appears more variable
- Weekday activity is more consistent

Recommendations

- Encourage moderate daily activity goals
- Provide sleep tracking insights tied to activity
- Use activity streaks to maintain engagement
- Bellabeat could introduce weekend engagement strategies, such as Saturday activity challenges or Sunday recovery reminders, to help users maintain consistent activity patterns across the week.
- User activity peaks on Saturday but drops nearly 15% on Sunday, suggesting an opportunity for engagement features that encourage users to maintain activity consistency at the end of the weekend.

Conclusion

- This analysis demonstrates how daily activity data can reveal patterns in physical behavior, energy expenditure, and sleep. These insights can help Bellabeat design more personalized wellness recommendations and engagement strategies.
- For questions and feedback please contact me through my Github Portfolio
- Github contains presentation, Python code, and dashboard link